



6 Months Certification Course

Syllabus 2026

Cyber Security

Prepared By:
TECH ENGINEER, Mohali

This Cyber Security program is designed to provide complete theoretical & practical knowledge of online and offline security tools and strategies. Students will learn how to be secure & safe from cyber frauds / crime and build successful careers.

PROGRAM IS BEST SUITED FOR



Entrepreneurs



College Students



Marketing Professionals



Job Seekers

WHY YOU SHOULD LEARN CYBER SECURITY

Learning cybersecurity is one of the most strategic moves you can make in the digital age, whether for professional growth or personal safety.

1. **High Demand:** Jobs for information security analysts are projected to grow 33% through 2033, far outstripping most other industries.
2. **Competitive Salaries:** Due to the skill shortage, salaries are among the highest in the IT sector. Entry-level roles can average around \$92,000–\$93,500 per year, while senior roles like Chief Information Security Officer (CISO) can exceed \$250,000–\$273,000.
3. **Variety of Specialties:** You aren't locked into one path. You can specialize in Ethical Hacking, Digital Forensics, Cloud Security, Risk Management, or Incident Response.
4. **Societal Contribution:** Your work directly prevents identity theft, financial fraud, and data breaches for millions of people.
5. **Protect Yourself:** Learning the basics empowers you to secure your own devices, passwords, and banking details against phishing and malware.
6. **Accessibility:** You don't always need a computer science degree to start. Many enter the field through certifications or self-study platforms like TryHackMe and Hack The Box.

By learning Cyber Security, you gain the ability to safe and secure from online frauds and cyber crimes, and will get functional digital experiences.

Module-I: Introduction to Cyber security

- Defining Cyberspace
- Overview of Computer and Web-technology
- Types of Hackers (White Hat, Black Hat, Grey Hat)
- Architecture of cyberspace
- Communication and web technology
- Internet, World wide web, Advent of internet,
- Internet infrastructure for data transfer and governance,
- Internet society & Regulation of cyberspace
- Basics of Ethical Hacking
- Concept of Cyber Security
- Issues and challenges of Cyber Security.

Learning Outcomes: After completion of this module, students would be able to understand the concept of Cyber security and issues / challenges.

Module-II: Cyber crime and Cyber law

- Classification of cyber crimes
- Common cyber crimes
- Cyber crime targeting computers and mobiles
- Cyber crime against women and children
- Financial frauds
- Social engineering attacks
- Malware and ransomware attacks
- Zero day and zero click attacks
- Cyber criminals modus-operandi
- Reporting of cyber crimes
- Remedial and mitigation measures
- Legal perspective of cyber crime
- IT Act 2000 and its amendments
- Cyber crime and offenses
- Organizations dealing with Cyber crime and Cyber security in India
- Case studies

Learning Outcomes: Students, at the end of this module, should be able to understand the cyber crimes, their nature, legal remedies and as to how report the crimes through available platforms and procedures.

Module-III: Social Media Overview and Security

- Introduction to Social networks
- Types of Social media
- Social media platforms
- Social media monitoring
- Hashtag and Viral content
- Social media marketing
- Social media privacy
- Challenges and Opportunities
- Social Media Security Issues
- Inappropriate content reporting
- Inappropriate content posting Laws
- Cyber Laws in India
- Social Media Best practices
- Case studies

Learning Outcomes: On completion of this module, students should be able to appreciate various privacy and security concerns on online Social media and understand the reporting procedure of inappropriate content, underlying legal aspects and best practices for the use of Social media platforms.

Module IV: E-Commerce and Digital Payments

- Definition of E- Commerce
- Main components of E-Commerce
- Elements of E-Commerce security
- E-Commerce threats
- E-Commerce security best practices
- Introduction to digital payments
- Components of digital payment and stake holders
- Modes of digital payments -
 - Banking Cards
 - Unified Payment Interface (UPI)
 - e-Wallets
 - Unstructured Supplementary Service Data (USSD)
 - Aadhar enabled payments
- Digital payments related common frauds and preventive measures.
- RBI guidelines on digital payments
- Customer protection in unauthorized banking transactions

Learning Outcomes: After the completion of this module, students would be able to understand the basic concepts related to E-Commerce and digital payments. They will become familiar with various digital payment modes and related cyber security aspects, RBI guidelines and preventive measures against digital payment frauds.

Module V: Digital Devices Security, Tools and Technologies

- End Point device and Mobile phone security
- Password policy
- Security patch management
- Data backup & Downloading
- Management of third party software
- Device security policy
- TCP/IP Protocol, IP Addressing & Subnetting
- DNS, DHCP, HTTP, HTTPS
- Cyber Security best practices
- Significance of host firewall and Ant-virus
- Management of host firewall and Anti-virus
- Wi-Fi security
- Configuration of basic security policy and permissions

Module VI: Operating System Security

- Windows Security Basics
- Linux Fundamentals
- User & Permission Management
- File System Security
- System Hardening
- Web Application Security
 - How websites work
 - HTTP Requests & Responses
 - Common Web Vulnerabilities
 - SQL Injection
 - Cross-Site Scripting (XSS)
 - Broken Authentication
 - Security Misconfiguration
- System Hacking
 - Password Cracking
 - Privilege Escalation
 - Malware Basics
 - Backdoors & Trojans
- Cloud Security
 - Cloud Fundamentals
 - Security in AWS / Azure
 - Identity & Access Management
 - Cloud Threats

Learning Outcomes: Students, after completion of this module will be able to understand the basic security aspects related to Computer and Mobiles. They will be able to use basic tools and technologies to protect their devices.



Certificate

Upon completing this Certification course, you will receive a program completion certificate from Tech Engineer, Mohali (Pb). This certificate will testify to your skills as an expert in the completed course.

Contact Us

Visit Our HQ



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